Electrical Circuit

沈威宇

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1 Electrical Circuit

I Introduction

i Convention

- Time t (s)
- Period T (s)
- Complex frequency (Laplace domain) s (Hz = 1 / s)
- Angular or radial frequency ω (Hz = 1 / s)
- Voltage v (time domain) **V** (phasor domain in peak/magnitude) $\mathbf{V}_{rms}$ (phasor domain in RMS) V (Laplace domain) (volt = V) across (跨壓)
- Current i (time domain) **I** (phasor domain) I (Laplace domain) (ampere = A) through
- Work w (J)
- Power p (watt = W = A V)
- Real, average, or active power P (W)
- Apparent power S (volt-ampere = VA)
- Complex power **S** (VA)
- Reactive power Q (volt-ampere-reactive = VAR)
- Imaginary unit (since i is used by current) $j = \sqrt{-1}$
- Resistance R (ohm = Ω= V / A)
- Capacitance C (farad = F = A s / V = S s)
- Inductance L (henry = H = V s / A = Ωs)
- Reactance X (Ω)
- Impedance Z (Ω)
- Admittance Y (siemens = S = 1 / Ω)
- Conductance G (S)
- Susceptance B (S)
- Time constant τ (s)
- Conjugate *
- Dirac delta function $\delta(t)$
- Unit step function $u(t)$
- Impulse response function $h(t)$

- Transfer function $H(s)$
- Phase (angle) φ:

$$\varphi = 2\pi \left(\frac{t - t_0}{T} - \left\lfloor \frac{t - t_0}{T} \right\rfloor \right).$$

ii Current

Current, $i = \frac{\mathrm{d}q}{\mathrm{d}t}$ has direction, and the current reference direction is defined as the direction of equivalent positive charge transfer.

iii Voltage

Voltage, $v = \frac{\mathrm{d}w}{\mathrm{d}q}$, has polarity, and the voltage reference polarity is + at the higher potential terminal and − at the lower potential terminal.

iv Electrical network

An electrical network or simply network is an interconnection of electrical components or a model of such an interconnection, consisting of electrical elements.

v Electrical element

Electrical elements or simply elements are conceptual abstractions representing idealized electrical components, such as resistors, capacitors, and inductors, used in the analysis of electrical networks and denoted with circuit (Ckt) symbols, also called electronic symbol, in circuit diagrams.

vi Average as Cesàro mean

The term average used below is defined as Cesàro mean over $[0, \infty)$.

vii Active element

An active element is an element capable of supplying a power with average over $[0, \infty)$ greater than zero to some external network.

viii Passive element

A passive element is an element that is not an active element.

ix Lumped-element model, lumped-parameter model, or lumped-component model

A simplified and idealized representation of an electrical network that assumes all components are concentrated at a single point and their behavior can be described by idealized mathematical models, that is, the lumped-matter discipline:

- The change of the magnetic flux in time through the surface surrounded by a loop is zero.

- The change of the charge in time inside conductor is zero.
- Signal timescales of interest are much larger than propagation delay of electromagnetic waves across the lumped element.

such that the system is a dynamic system whose state space is a finite dimension, called lumped-parameter system.

In contrast, distributed parameter systems have infinite-dimensional state space.

We will concentrate on lumped-parameter networks in this text.

x Time-invariant network

Elements' properties and connections don't change over time. We will concentrate on time-invariant networks in this text.

xi Circuit element

A circuit element is a two terminal electrical network that can be completely characterized by its voltage-current relationship.

We label v and parity for voltage and i and direction for current. If the current arrow enters the + terminal, the element is said to follow the passive (polarity/sign) convention. Under this convention, the power absorbed (dissipated) by the element is

$$p(t) = v(t)i(t),$$

with negative power means supplying. This convention is used for passive elements.

If the current arrow enters the - terminal, the element is said to follow the active (polarity/sign) convention. Under this convention, the power supplied (generated) by the element is

$$p(t) = v(t)i(t),$$

with negative power means absorbing. This convention is used for active elements.

xii Direct current (DC)

A direct current is a current that is one-directional and with constant voltage and current over time.

xiii General current and Laplace domain

Given a general current with voltage $v(t)$ and current $i(t)$ over time since $t = 0$, we can perform Laplace transforms,

$$V(s) = \mathcal{L}\{v(t)\}(s) = \int_0^\infty e^{-st} v(t)\,\mathrm{d}t\,,$$

$$I(s) = \mathcal{L}\{i(t)\}(s) = \int_0^\infty e^{-st} i(t)\,\mathrm{d}t\,,$$

to obtain its voltage $V(s)$ and current $I(s)$ as a function of Laplace domain s.

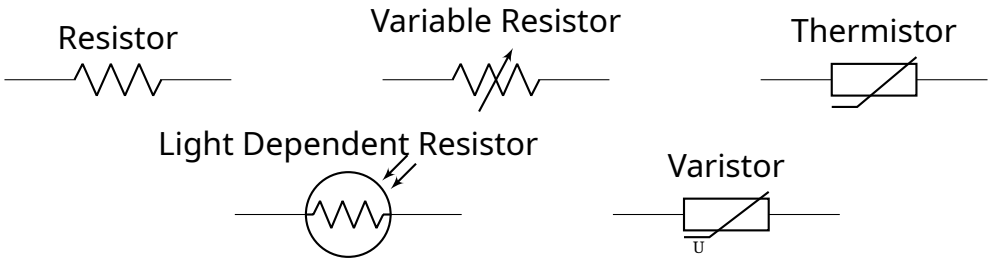
xiv Resistance and resistor

The resistance R of an element is the voltage across it over the current through it under DC:

$$v(t) = Ri(t).$$

$$V(s) = RI(s).$$

An ideal resistor is a circuit element with a resistance and no capacitance or inductance.



Power absorbed:

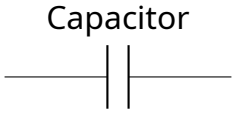
$$p(t) = \frac{(v(t))^2}{R} = (i(t))^2 R.$$

xv Potentiometer or pot and variable resistor or rheostat

A potentiometer or pot is a variable resistor with three terminals A, W, B in order with W called wiper. The resistance between A and B is constant and equals the sum of the resistance between A and W and the resistance between W and B, and the resistance between A and W (and thus W and B) can be adjusted manually by typically turning a knob or moving a slide. When only W and another terminal is used, it's usually called variable resistor or rheostat. It is commonly used to control voltage in a circuit.

xvi Capacitance and capacitor

An ideal capacitor is a circuit element with a capacitance and no resistance or inductance.



The capacitance C of an element is the current through it divided by the derivative of the voltage across it with respect to time, in which voltage must not be allowed to jump instantaneously from one value to another.

Time domain:

$$i(t) = C\frac{\mathrm{d}v(t)}{\mathrm{d}t}.$$

Laplace domain:

$$I(s) = CsV(s) - Cv(0^-).$$

$$V(s) = \frac{I(s)}{Cs} + \frac{v(0^-)}{s}.$$

A capacitor with capacitance C and initial voltage $v(0^-)$ is equivalent to a capacitance C without initial energy in parallel connection with an independent current source enforcing current

$-Cv(0^-)\delta(t)$ (time domain), i.e., $-Cv(0^-)$ (Laplace domain), where the real current through the capacitor is the sum of that of the two, or a capacitance C without initial energy in series connection with an independent voltage source enforcing voltage $v(0^-)u(t)$ (time domain), i.e., $\frac{v(0^-)}{s}$ (Laplace domain), where the real voltage across the capacitor is the sum of that of the two.

Integral voltage-current relationship:

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,\mathrm{d}\tau + v(t_0{}^-) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,\mathrm{d}\tau\,.$$

Power absorbed:

$$p(t) = i(t)v(t) = Cv(t)\frac{\mathrm{d}v(t)}{\mathrm{d}t}.$$

Energy stored:

$$\int_{-\infty}^{t} p(t)\,\mathrm{d}t = \frac{1}{2}C\,(v(t))^2\,.$$

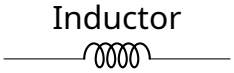
Charge in it:

$$q = CV$$

xvii Inductance and inductor

An ideal inductor is a circuit element with an inductance and no resistance or capacitance.

Inductor



The inductance L of an element is the voltage across it divided by the derivative of the current through it with respect to time, in which the current must not be allowed to jump instantaneously from one value to another.

Time domain:

$$v(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t}.$$

Laplace domain:

$$V(s) = LsI(s) - Li(0^-).$$
$$I(s) = \frac{V(s)}{Ls} + \frac{i(0^-)}{s}.$$

An inductor with inductance L and initial current $i(0^-)$ is equivalent to a inductance L without initial energy in series connection with an independent voltage source enforcing voltage $-Li(0^-)\delta(t)$ (time domain), i.e., $-Li(0^-)$ (Laplace domain), where the real voltage across the inductor is the sum of that of the two, or a inductance L without initial energy in parallel connection with an independent current source enforcing current $i(0^-)u(t)$ (time domain), i.e., $\frac{i(0^-)}{s}$ (Laplace domain), where the real current through the inductor is the sum of that of the two.

Integral voltage-current relationship:

$$i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,\mathrm{d}\tau + i(t_0{}^-) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,\mathrm{d}\tau\,.$$

Power absorbed:

$$p(t) = i(t)v(t) = Li(t)\frac{\mathrm{d}i(t)}{\mathrm{d}t}.$$

Energy stored:

$$\int_{-\infty}^{t} p(t)\,\mathrm{d}t = \frac{1}{2}L\left(i(t)\right)^2.$$

Flux linkage in it:

$$\lambda = LI$$

An ideal inductor has zero resistance. However, the wire used to wind the inductor's coil has resistance, which makes an inductor with positive resistance, called winding resistance.

Inductor is seldom used in microelectronics due to its large size, expensive price, and winding resistance.

xviii Resistive element

A resistive element is an element with nonzero resistance.

xix Reactive element

A reactive element is an element with either nonzero capacitance or nonzero inductance.

xx Purely resistive element

A pure resistive element is a resistive element with zero capacitance and zero inductance.

xxi Purely reactive element

A pure reactive element is a reactive element with zero resistance.

xxii Impedance (Laplace domain)

The impedance $Z(s)$ of a one-port network is defined as:

$$V(s) = Z(s)I(s),$$

and is such that for a one-port network with capacitance C:

$$Z(s) = \frac{1}{Cs};$$

and for a one-port network with inductance L:

$$Z(s) = Ls.$$

xxiii Admittance (Laplace domain)

The admittance $Y(s)$ of a one-port network is defined as the reciprocal of its impedance $Z(s)$.

xxiv Dynamic circuit

A circuit containing capacitance or inductance.

xxv Node

A node is a point in a network where two or more elements are connected.

xxvi Supernode

A supernode is any closed region in a network that contains two or more nodes whose boundary intersects some connecting wires of which each wire is intersected exactly once.

xxvii Pole or terminal

A pole or a terminal is a node that is available for connection to an external network.

xxviii Path

A path is any route through which current can flow from one node to another.

xxix Branch

A branch is a single element or a group of elements in series that connects two nodes. Branch variables are voltages or currents associated with individual branch.

xxx Loop

A loop is any closed path in a network where you can start at a node, traverse branches, and return to the starting node without retracing any branch.

xxxi (Essential) mesh

An (essential) mesh is a loop ℓ such that there is no other loop whose surface is a subset of the surface surrounded by ℓ.

xxxii Electrical circuit or closed circuit

An electrical circuit is a network containing loops.

xxxiii Open circuit (開路)

An open circuit is a path with infinite impedance. Open circuit voltage is denoted as v_{oc}.

xxxiv Short circuit (短路)

A short circuit is a path with zero impedance. Short circuit current is denoted as i_{sc}.

xxxv Port (埠)

A port is a pair of terminals through which current can enter or leave a network and satisfies the port condition: the current flowing into one pole from outside is equal to the current flowing out of the other pole to outside.

An n-port network is a network with $2n$ poles forming n ports.

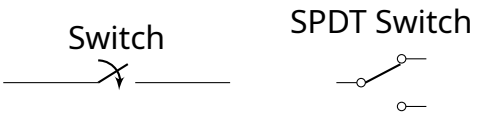
xxxvi Wire or lead

Circuit elements in a network are connected with wires (aka leads), which are assumed to have zero impedance.



Adding a wire directly between two nodes is called adding a short circuit.

xxxvii Switches



xxxviii Kirchhoff's current law (KCL), Kirchhoff's first law, or Kirchhoff's junction rule

The algebraic sum of the currents entering any node is zero.

xxxix Kirchhoff's voltage law (KVL), Kirchhoff's second law, or Kirchhoff's loop rule

The algebraic sum of the voltages around any closed path is zero.

xl Planar network

A network that is planar as a graph with nodes being vertices and branches as edges.

xli Series connection

Two or more elements are connected in series if they are connected along a single path, and each component has the same current through it, equal to the current through the network. The voltage across the network is equal to the sum of the voltages across each element. The current through the network and the current through each element is the same.

xlii Parallel connection

Two or more elements are connected in parallel if they are connected along multiple paths from one node to another, and each component has the same voltage across it, equal to the voltage through the network. The current through the network is equal to the sum of the current through each element. The voltage across the network and the voltage across each element is the same.

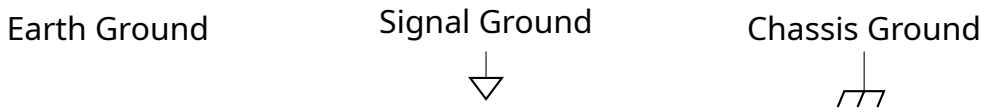
xliii Bilateral network

A bilateral electrical network is an electrical network whose behaviors remain the same regardless of the direction of current flow or applied voltage.

xliv Ground

A ground is a node designated to have 0 electric potential. All other electric potential in the network are measured relative to the grounds.

- **Earth ground**: Earth ground is a point in the network that is physically connected to the Earth.
- **Signal ground**: Signal ground is a ground in the network provided by an external signal, which can be float relative to Earth and such circuit is called float. An example of signal ground is a GND pin on a microcontroller board.
- **Chassis ground**: Chassis ground is a point in the circuit that is physically connected to a devices' metal frame. The metal frame acts as a shield against electromagnetic interference and can be connected to Earth for safety in AC-powered devices.



xlv Floating

A nodal voltage is floating with respect to another node if they do not share a common ground.

xlvi Operating point, Q-point, quiescent point, or bias point

The steady-state DC voltage and current values of a circuit, called biasing circuit, when no input signal is applied.

xlvii Biasing resistor

A biasing resistor is a resistor used to achieve a desired operating point.

xlviii Ladder network

A network whose circuit diagram forms a ladder.

xlix Time-invariant circuit

A circuit whose outputs as a function of the inputs is time-invariant. All circuits discussed here are time-invariant circuit.

l Input or excitation

External input to a network.

li Response

Any branch variable.

liii SPICE

SPICE (Simulation Program with Integrated Circuit Emphasis) is a general-purpose, open-source analog electronic circuit simulator. It is a program used in integrated circuit and board-level design to check the integrity of circuit designs and to predict circuit behavior.

liv Overcurrent or excess current

A situation where a larger than intended electric current exists.

lv Shunt

A shunt is a device designed to provide a low-impedance path for an electrical current in a circuit. It is typically used to divert current away from a system or component to, for example, prevent overcurrent with a shunt resistor or decrease high-frequency gain with a shunt capacitor. Connecting a node to ground with a shunt is called shunting the node to ground.

II Source

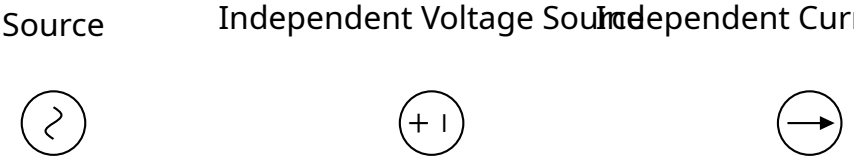
i Ideal source

Ideal voltage sources are active elements with zero output resistance. To turn off an ideal voltage source means replace it with short circuit.

Ideal current sources are active elements with infinite output resistance. To turn off an ideal current source means replace it with open circuit.

ii Independent source

- An independent voltage source enforces a prescribed voltage v_s across its terminals.
- An independent current source enforces a prescribed current i_s through its terminals.



iii Linear dependent or controlled source

Control voltages and currents are called control variables.

- A voltage-controlled voltage source (VCVS) enforces voltage $v_s = \mu v_x$ across its terminals, where v_x is control voltage and μ is dimensionless voltage gain.

- A current-controlled voltage source (CCVS) enforces voltage $v_s = ri_x$ across its terminals, where i_x is control current and r is transresistance.
- A voltage-controlled current source (VCCS) enforces current $i_s = gv_x$ through its terminals, where v_x is control voltage and g is transconductance.
- A current-controlled current source (CCCS) enforces current $i_s = \beta i_x$ through its terminals, where i_x is control current and β is dimensionless current gain.

Controlled Voltage Source Controlled Current Source



iv Load network (負載網路)

A two-terminal network containing no independent sources and no controlled sources whose control variable is not within the network.

v Equivalent resistance and impedance theorem

For a load network consisting entirely of resistances and controlled sources, the terminal voltage and current can be related by

$$v = R_{eq} i,$$

where R_{eq} is called equivalent resistance.

For a load network consisting entirely of impedances and controlled sources, the terminal voltage and current in Laplace domain can be related by

$$V = Z_{eq} I,$$

where Z_{eq} is called equivalent impedance.

vi Real source

A real independent voltage source can be modeled with an ideal independent voltage source in series connection with a resistor.

A real independent current source can be modeled with an ideal independent current source in parallel connection with a resistor.

III Linear time-invariant (LTI) circuit

i Linear element

A linear element is an element whose inputs $\mathbf{x}$ and outputs $\mathbf{F}(\mathbf{x})$ satisfy linearity:

$$\mathbf{F}(a\mathbf{x} + b\mathbf{y}) = a\mathbf{F}(\mathbf{x}) + b\mathbf{F}(\mathbf{y})$$

such as an impedance and a linearly controlled source.

$$\mathbf{F}(a\mathbf{x}) = a\mathbf{F}(\mathbf{x})$$

is called proportionality principle/property and

$$\mathbf{F}(\mathbf{x}+\mathbf{y})=\mathbf{F}(\mathbf{x})+\mathbf{F}(\mathbf{y})$$

is called superposition principle/property.

ii Linear network

A linear network is a network whose inputs $\mathbf{x}$ and outputs $\mathbf{F}(\mathbf{x})$ satisfy linearity:

$$\mathbf{F}(a\mathbf{x}+b\mathbf{y})=a\mathbf{F}(\mathbf{x})+b\mathbf{F}(\mathbf{y})$$

Equivalently, a linear network is a network consisting only of linear elements and independent sources.

iii Linear circuit

A linear circuit is a circuit consisting only of linear elements and independent sources.

iv Series connection

For linear networks, the impedance, reciprocal of capacitance, and inductance of the network are equal to the sums of those of each element. When steady, the charges in each capacitor and the equivalent capacitor are the same.

v Parallel connection

For linear networks, the admittance, capacitance, and reciprocal of inductance of the network are equal to the sums of those of each element. When steady, the flux linkage in each inductor and the equivalent inductor are the same.

vi Linear circuit as graph

A linear circuit is a graph with nodes with all voltage sources enclosed as supernodes being vertices, branches without nodes inside except ends being edges, and admittances (or conductance for DC) being weights.

vii decibel (dB)

For voltage or current, magnitude response, input, and output can be expressed in decibel (dB) as $g(\omega)\coloneqq 20\log_{10}|H(i\omega)|$, $20\log_{10}|x|$, and $20\log_{10}|y|$.

For power, magnitude response, input, and output can be expressed in decibel (dB) as $g(\omega)\coloneqq 10\log_{10}|H(i\omega)|$, $10\log_{10}|x|$, and $10\log_{10}|y|$.

IV Thévenin's theorem and Norton's theorem

i Thévenin's theorem

Any linear network with two terminals is equivalent to a voltage source with a voltage called Thévenin voltage in a series connection with a linear element (or resistor for DC) with

an impedance (or resistance for DC) called Thévenin, series, or equivalent impedance (or resistance for DC), such equivalent circuit is called Thévenin equivalent.

ii Norton's theorem

Any linear network with two terminals is equivalent to a current source with a current called Norton current in a parallel connection with a linear element (or resistor for DC) with an impedance (or resistance for DC) called Norton, parallel, or equivalent impedance (or resistance for DC), such equivalent circuit is called Norton equivalent.

iii Finding Thévenin voltage

Remove all loads, and the open-circuit voltage v_{oc} between the two terminals is the Thévenin voltage.

iv Finding Norton current

Add a short circuit between the two terminals, and the short circuit current i_{sc} on it is the Norton current.

v Finding Thévenin impedance or Norton impedance

Thévenin and Norton impedance Z_{th} (or resistance R_{th} for DC) are the same and can be found with:

- **Source deactivation method**: Replace all independent voltage sources with short circuits and independent current sources with open circuits,
  - If there are no dependent sources, the equivalent impedance between the two terminals of the dead network is the Thévenin impedance.
  - **Driving-point impedance method or test source method**: Otherwise, put an arbitrary nonzero-valued independent voltage V_t or current I_t source between the two terminals, and the voltage divided by the current of the added source $\frac{V_t}{I_t}$, called driving-point impedance, is the Thévenin impedance.
- **Division method**: The Thévenin impedance or Norton impedance is the Thévenin voltage divided by the Norton current.

vi Thévenin parameters

Thévenin voltage, Norton current, and Thévenin impedance are called Thévenin parameters.

vii Maximum power theorem under DC

A one-port linear network delivers maximum power to a load resistance when the load resistance is equal to the Thévenin resistance R_{th} of the network, and the power delivered

is

$$P = \frac{{V_{oc}}^2}{4R_{th}}.$$

V Nodal/node analysis or node-voltage analysis of linear circuits

i Nodal analysis

List KCL of each node and solve them.

ii (Nodal) admittance and conductance matrix

The (nodal) admittance (conductance for DC) matrix **Y** (**G** for DC) of a linear circuit is the Laplacian matrix of the circuit as weighted undirected graph with all sources omitted and equivalent admittances (conductances for DC) as weights and nodes not connected as connected with zero admittance (conductance for DC) and thus 0 weight, that is, a matrix of number of nodes rows and columns with Y_{ii} being the total equivalent admittances (conductances for DC) connecting node i and adjacent nodes with branches with sources omitted and Y_{ij} for $i \neq j$ being 0 if nodes i and j are not connected and the negation of the equivalent admittance between them with branches with sources omitted otherwise.

iii Nodal matrix equation and modified nodal analysis (MNA)

For a linear circuit,

1. Choose a reference node and assign a voltage for it (typically 0 and denoted ground symbol).
2. For each voltage source whose either endpoint is the reference node, the voltage of the other is known.
3. For each pair of node with a voltage source in series connection with an nonzero impedance (resistance for DC) connecting them, convert it to its Norton equivalent if for normal nodal analysis.
4. For each node with unknown voltage, assign a nodal voltage variable.
5. Define $\mathcal{V}$ as the column vector of the nodal voltage variables.
6. Define $\mathcal{I}$ as the column vector of, for each node k, the sum of
   - currents injected into node k from current sources, and
   - voltages of adjacent nodes j of known nodal voltages multiplied by the equivalent admittance (conductance for DC) between nodes j and k, that is, the current injected into node k from node j if node k is ground,

   all with control functions used when involving controlled sources.
7. Define **Y** as the (nodal) admittance (conductance for DC) matrix of the circuit. Note that nodes with known nodal voltage are not in $\mathcal{V}$ but admittances of edges to them still count when building a_{ii} of **Y**.

8. If there isn't voltage source whose neither endpoint is the reference node,

   (a) Solve the system of linear equations:

$$\mathbf{Y}\mathcal{V} = \mathcal{I}.$$

   Otherwise, current through such voltage source from positive terminal to negative terminal is called fictitious source current, the analysis is called modified nodal analysis (MNA), and each kth such source are referred to as source k below.

   (a) Define the kth column of the matrix $\mathbf{A}_v$ as 1 for ith row such that source k's positive terminal is node i, -1 for jth row such that source k's negative terminal is node j, and 0 for all other rows.

   (b) Define the kth row of the column vector $\mathbf{V}_s$ as source k's voltage.

   (c) Assign current variable i_k as the fictitious current of source k, and define the kth row of the column vector $\mathcal{I}_v$ as i_k.

   (d) Solve the system of linear equations:

$$\begin{bmatrix} \mathbf{Y} & \mathbf{A}_v \\ \mathbf{A}_v^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathcal{V} \\ \mathcal{I}_v \end{bmatrix} = \begin{bmatrix} \mathcal{I} \\ \mathbf{V}_s \end{bmatrix}.$$

VI (Matrix) mesh analysis of planar linear circuit

i (Mesh) impedance matrix

The (mesh) impedance (resistance for DC) matrix **Z** of a linear circuit that is a simple planar graph is the matrix with entry Z_{ii} being the sum of the impedances of the branches in the ith mesh and Z_{ij} for $i \neq j$ being the negation of the sum of the impedances of the branches shared between the ith and jth meshes.

ii Mesh analysis

1. For each pair of node with a current source in parallel connection with an nonzero impedance (resistance for DC) connecting them, optionally convert it to its Thévenin equivalent.
2. For meshes with current source on it and not on any other mesh, the current is known.
3. Assign an arbitrary but consistent direction (usually clockwise) of current for all other meshes, and assign a mesh current variable for each mesh.
4. For each current source that lies on two meshes, called interior current source, exclude the edge with the current source, and merge the two meshes into one mesh, called supermesh.
5. Suppress all current sources. Solve the system of linear equations of KVL of the meshes.

iii Mesh matrix equation and modified kesh analysis (MMA)

1. Construct mesh and assign variables the same as mesh analysis except that interior current sources are preserved and supermeshes are not formed.
2. Define $\mathcal{I}$ as the column vector of the mesh current variables.
3. Define $\mathcal{V}$ as the column vector of the sum of
   - voltage delivered from the voltage sources in the mesh with mesh current variable direction as positive direction, and
   - current of adjacent meshes with known current with same direction in shared branch as negative multiplied by the impedance of the shared branch,

   all with control functions used when involving controlled sources.
4. Define **Z** as the (mesh) impedance (resistance for DC) matrix of the circuit.
5. For each current source, append the equation to the system with the enforced current under $\mathcal{V}$ and the row vector of weights of mesh current variables under **Z**.
6. If there is no interior current source,

   (a) Solve the system of linear equations:

   $$\mathbf{Z}\mathcal{I} = \mathcal{V}.$$

   Otherwise, voltage across such current source of input terminal relative to output terminal is called fictitious source voltage, the analysis is called modified mesh analysis (MMA), and each kth such source are referred to as source k below.

   (a) Define the kth column of the matrix $\mathbf{A}_c$ as 1 for ith mesh such that source k's is on mesh i and of the same current direction, -1 for jth row such that source k is on mesh j and of opposite current direction, and 0 for all other rows.

   (b) Define the kth row of the column vector $\mathbf{I}_s$ as source k's current.

   (c) Assign voltage variable v_k as the fictitious voltage of source k, and define the kth row of the column vector $\mathcal{V}_c$ as v_k.

   (d) Solve the system of linear equations:

   $$\begin{bmatrix} \mathbf{Z} & \mathbf{A}_i \\ \mathbf{A}_i^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathcal{I} \\ \mathcal{V}_c \end{bmatrix} = \begin{bmatrix} \mathcal{V} \\ \mathbf{I}_s \end{bmatrix}.$$

VII Transformations

i Source transformation

An independent voltage source V in series connection with an impedance Z is equivalent to an independent current source $I = \dfrac{V}{Z}$ in parallel connection with the same impedance Z.

ii Y-Δ transformation and Δ-Y transformation

- Y (wye or star) network: a common central node N, three terminals N_1, N_2, N_3, and three circuit elements with impedances Z_1, Z_2, Z_3 connecting N_1 and N, N_2 and N, and N_3 and N respectively.
- Δ (delta or triangle) network: three terminals N_1, N_2, N_3, and three circuit elements with impedances Z_{12}, Z_{23}, Z_{31} connecting N_1 and N_2, N_2 and N_3, and N_3 and N_1 respectively.
- Y-Δ transformation: The two networks are equivalent if

$$Z_{12} = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3},$$
$$Z_{23} = Z_2 + Z_3 + \frac{Z_2 Z_3}{Z_1},$$
$$Z_{31} = Z_3 + Z_1 + \frac{Z_3 Z_1}{Z_2}.$$

- Δ-Y transformation: The two networks are equivalent if

$$Z_1 = \frac{Z_{12} Z_{31}}{Z_{12} + Z_{23} + Z_{31}},$$
$$Z_2 = \frac{Z_{23} Z_{12}}{Z_{12} + Z_{23} + Z_{31}},$$
$$Z_3 = \frac{Z_{31} Z_{23}}{Z_{12} + Z_{23} + Z_{31}}.$$

iii Star-mesh transformation

For all integer $n \geq 3$:

- n-star network: a common central node N, n terminals $N_1, N_2, \dots, N_n$, and n circuit elements with impedances Z_i connecting N_i and N for all $i \in \mathbb{N} \wedge i \leq n$.
- n-mesh (complete graph) network: n terminals $N_1, N_2, \dots, N_n$, and $\frac{n(n-1)}{2}$ circuit elements with impedances Z_{ij} connecting N_i and N_j for all $i, j \in \mathbb{N} \wedge i < j \leq n$.
- Star-mesh transformation: The two networks are equivalent if

$$Z_{ij} = Z_i + Z_j + \frac{Z_i Z_j}{-Z_i - Z_j + \sum_{k=1}^{n} Z_k}.$$

VIII Responses of LTI circuits

i Natural, homogeneous, unforced, or zero-input response (ZIR)

The response with all independent sources in a LTI circuit turned off and initial states preserved.

The ZIR is the solution of an IVP of a linear homogeneous ODE, called governing equation, with initial states given by the initial states of inductors and capacitors.

Circuit whose governing equation is an nth-order differential equation is called an nth-order circuit.

ii Forced, driven, particular, or zero-state responses (ZSR)

The response with only one of the independent sources in the LTI circuit, called forcing function, turned on, and with zero initial state.

Each ZSR is the response of a causal continuous-time linear time-invariant (LTI) systems with its forcing function as forcing function and the complementary equation of its governing equation being the governing equation of ZIR.

iii Total or complete response

The response of the original circuit.

iv Superposition principle/theorem

The sum of ZIR and all ZSRs is the total response.

v Step response

ZSR of step source, that is, an independent source whose enforced voltage or current is a multiple of the step function $u(t)$.

vi Duality

Two linear circuit are dual if their governing equations in terms of variables become identical after systematically exchanging voltage and current, capacitance and inductance, impedance and admittance, open circuit and short circuit, KCL and KVL, Thévenin's theorem and Norton's theorem, series and parallel, node and loop, etc.

Two linear circuit are exact dual if their governing equations in terms of numerical values become identical after systematically exchanging voltage and current, capacitance and inductance, impedance and admittance, open circuit and short circuit, KCL and KVL, Thévenin's theorem and Norton's theorem, series and parallel, node and loop, etc.

vii Quality factor (品質因數) or Q factor

The quality factor or Q factor Q of a network is 2π multiplied by the maximum energy stored in a period, E_m, divided by the total energy lost in the period, E_l, during natural response, that is,

$$Q = \frac{2\pi E_m}{E_l}.$$

IX Steady-state sinusoidal AC

i Effective value or root mean square (RMS)

The effective value or root mean square (RMS) of a voltage or current is defined as its RMS over $[0, \infty)$.

ii Steady-state sinusoidal alternating current (AC) and phasor domain

A steady-state sinusoidal alternating current (AC), often simply called alternating current (AC), is a current whose voltage $v(t)$ and current $i(t)$ are of the form:

$$v(t) = v_m \cos(\omega t + \varphi),$$

$$i(t) = i_m \cos(\omega t + \theta),$$

where ω is the angular frequency of them.

The period T and frequency f of them are

$$T = \frac{2\pi}{\omega},$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi}.$$

The RMSs are

$$v_{rms} = \sqrt{\frac{1}{T}\int_0^T (v(t))^2 \,\mathrm{d}t} = \frac{v_m}{\sqrt{2}},$$

$$i_{rms} = \sqrt{\frac{1}{T}\int_0^T (i(t))^2 \,\mathrm{d}t} = \frac{i_m}{\sqrt{2}}.$$

They can be represented in phasors in peak/magnitude as

$$\mathbf{V} = v_m \angle\varphi = v_m e^{j\varphi},$$

$$\mathbf{I} = i_m \angle\theta = i_m e^{j\theta},$$

or phasors in RMS as

$$\mathbf{V}_{rms} = v_{rms} \angle\varphi = v_{rms} e^{j\varphi},$$

$$\mathbf{I}_{rms} = i_{rms} \angle\theta = i_{rms} e^{j\theta},$$

where the later are used in power analysis and the former are used in most other scenario.

Phasor domain is also called frequency domain when not confused with Laplace domain.

Lagging and leading mean current relative to voltage unless otherwise specified.

iii Capacitor under steady-state sinusoidal AC

Governing equation:

$$\mathbf{I} = Cj\omega\mathbf{V}.$$

Response: If

$$v(t) = v_m \cos(\omega t + \theta),$$

$$\mathbf{V} = v_m e^{j\theta},$$

then

$$i(t) = C\omega v_m \cos\left(\omega t + \theta + \frac{\pi}{2}\right),$$

$$\mathbf{I} = C\omega v_m e^{j(\theta + \frac{\pi}{2})}.$$

The current leads voltage by $\frac{\pi}{2}$, and the higher angular frequency ω is, the lower $\frac{v_m}{i_m} = \frac{1}{C\omega}$ is.

iv Inductor under steady-state sinusoidal AC

Governing equation:

$$\mathbf{V} = Lj\omega\mathbf{I}.$$

Response: If

$$i(t) = i_m \cos(\omega t + \theta),$$

$$\mathbf{I} = i_m e^{j\theta},$$

then

$$v(t) = L\omega i_m \cos\left(\omega t + \theta + \frac{\pi}{2}\right),$$

$$\mathbf{V} = L\omega i_m e^{j(\theta+\frac{\pi}{2})}.$$

The current lags voltage by $\frac{\pi}{2}$, and the higher angular frequency ω is, the higher $\frac{v_m}{i_m} = L\omega$ is.

v Reactance (phasor domain)

The (capacitive) reactance X_C of a one-port network with capacitance C under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X_C = -\frac{1}{C\omega}.$$

The (inductive) reactance X_L of a one-port network with inductance L under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X_L = L\omega.$$

The reactance X of a one-port network under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X = X_L + X_C.$$

Another choice is to define X_C as $\frac{1}{C\omega}$ and $X = X_L - X_C$.

vi Impedance (phasor domain)

The impedance Z of a one-port network under steady-state sinusoidal AC of angular frequency ω is defined as:

$$\mathbf{V} = Z\mathbf{I},$$

and is such that

$$Z = R + jX.$$

The triangle on the complex plane of Z, R, and jX showing

$$Z = R + jX$$

is called the impedance triangle.

The impedance in phasor domain can be obtained by assigning $s = j\omega$ to the impedance in Laplace domain.

vii Admittance (phasor domain)

The admittance Y of a one-port network is defined as the reciprocal of its impedance Z.

viii Conductance (phasor domain)

The conductance G of a circuit element is defined as the real part of its admittance.

ix Susceptance (phasor domain)

The susceptance B of a circuit element is defined as the imaginary part of its admittance.

X Steady-state sinusoidal AC power analysis

i Instantaneous power

$$\begin{aligned} p(t) = v(t)i(t) &= v_m i_m \cos(\omega t + \theta)\cos(\omega t + \theta - \varphi) \\ &= v_{rms} i_{rms} \left(\cos\varphi + \cos(2\omega t + 2\theta - \varphi)\right) \end{aligned}$$

ii Complex power

The complex power **S** with unit volt-ampere (VA) of a steady-state sinusoidal AC is defined as

$$\mathbf{S} = \mathbf{V}_{rms}\mathbf{I}_{rms}{}^{*} = v_{rms} i_{rms} e^{j\varphi}.$$

iii Power conservation law for complex power

The sum of complex power from all elements in a linear circuit under steady-state AC is 0.

iv Real, average, or active power

The real, average, or active power P with unit watt (W) of a steady-state sinusoidal AC is defined as the average power of it, that is,

$$P = \langle p(t) \rangle = v_{rms} i_{rms} \cos\varphi = \Re(\mathbf{S}).$$

Average power of any pure reactive one-port network is 0.

v Apparent power

The apparent power S with unit volt-ampere (VA) of a steady-state sinusoidal AC is defined as

$$S = |\mathbf{S}| = v_{rms} i_{rms}.$$

vi Power factor (PF, pf, or P.F.)

The power factor (PF) of a steady-state sinusoidal AC is defined as

$$\text{PF} = \frac{P}{S} = \cos\varphi.$$

PF = 1 is called ideal. We also often write lagging or leading after power factor.

vii Reactive power

The reactive power Q with unit volt-ampere-reactive (VAR or VAr) of a steady-state sinusoidal AC is defined as

$$Q = \mathfrak{I}(\mathbf{S}) = v_{rms} i_{rms} \sin\varphi,$$

which is positive for lagging and negative for leading. It's the power that oscillates between sources and reactive elements.

viii Power triangle

The triangle on the complex plane of $\mathbf{S}$, P, and jQ showing

$$\mathbf{S} = P + jQ$$

is called the power triangle.

ix Source power factor

A source's power factor is in $[-1, 1]$, and is ± 1 iff the sum of Q of all other elements is 0, and 0 iff the sum of P of all other elements is 0.

x Maximum power theorem under steady-state sinusoidal AC

A one-port linear network delivers maximum power to a load impedance when the load impedance is equal to the conjugate of the Thévenin impedance Z_{th} of the network, and the average power delivered is

$$P = \frac{\left|\mathbf{V}_{oc}\right|^2}{8\Re(Z_{th})} = \frac{\left|\mathbf{V}_{oc,rms}\right|^2}{4\Re(Z_{th})}.$$

xi Wattmeter

A wattmeter measuring a load Z contains a current-sensing coil with current $i_i(t)$ in series connection with the load, and a voltage-sensing coil with current $i_v(t)$ in series connection with an impedance Z_m, in parallel connection with the load. Electrodynamic interaction produce a torque proportional to $i_v(t)i_i(t)$ on the rotating pointor. The steady-state deflection angle γ of the rotating pointor is proportional to the average of $i_v(t)i_i(t)$:

$$\gamma = \frac{K_M}{T} \int_{t_1}^{t_1+T} i_v(t) i_i(t) \,\mathrm{d}t\,,$$

where K_M is a proportionality constant and T is the period of the current. The impedance Z_m is a large resistance $R_m \gg |Z|$. If the load has

$$v(t) = \sqrt{2} v_{rms} \cos(\omega t + \theta)$$

$$i(t) = \sqrt{2} i_{rms} \cos(\omega t + \theta - \varphi),$$

then

$$i_v(t) = \frac{\sqrt{2} v_{rms}}{R_M} \cos(\omega t + \theta)$$

$$i_i(t) \approx \sqrt{2} i_{rms} \cos(\omega t + \theta - \varphi).$$

Then,

$$\gamma \approx \frac{K_M}{R_M} v_{rms} i_{rms} \cos\varphi = \frac{K_M}{R_M} P,$$

and reads approximately P.

An ideal wattmeter reads exactly P and $i_v \to 0$.

By convention, the current input terminal is labeled $\pm$.

xii Balanced three-phase circuit

A power system supplied by just one sinusoidal AC source is called single-phase (1-φ). The oscillatory behavior of instantaneous power puts pulsating strain on the generating and load equipments. Polyphase circuit with multiple sources were developed in response to this problem. The most common one us the balanced three-phase (3-φ) circuit, which has three sources with $\frac{2\pi}{3}$ phase difference and has constant instantaneous power.

The equation

$$v_a(t) = \sqrt{2} v_{rms} \cos(\omega t + \theta)$$
$$v_b(t) = \sqrt{2} v_{rms} \cos\left(\omega t + \theta - \frac{2\pi}{3}\right)$$
$$v_c(t) = \sqrt{2} v_{rms} \cos\left(\omega t + \theta + \frac{2\pi}{3}\right)$$

defines a symmetrical three-phase set with phase sequence $a-b-c$, where the phase sequence indicates the time sequence of the waveform peaks. Their sum is zero.

A (balanced) wye (Y) generator consists of a node called neutral and three branches with phase voltages $\mathbf{V}_{rms,a}, \mathbf{V}_{rms,b}, \mathbf{V}_{rms,c}$ w.r.t. the neutral that have phase difference $\frac{2\pi}{3}$ among each other, and line voltages $\mathbf{V}_{rms,ab}, \mathbf{V}_{rms,bc}, \mathbf{V}_{rms,ca}$ are defined across each pairs of their terminals.

$$\mathbf{V}_{rms,ab} = \mathbf{V}_{rms,a} - \mathbf{V}_{rms,b}$$
$$\mathbf{V}_{rms,bc} = \mathbf{V}_{rms,b} - \mathbf{V}_{rms,c}$$
$$\mathbf{V}_{rms,ca} = \mathbf{V}_{rms,c} - \mathbf{V}_{rms,a}$$

An equivalent generator is (balanced) delta (Δ) generator obtained by applying Y-Δ transformation to the Y generator, in which the neutral and phase voltages do not physically exist.

Similarly, a (balanced) wye load has a neutral and three branches with identical phase impedances, and a (balanced) delta (Δ) load is it applied Y-Δ transformation.

A three-phase circuit with wye generator and loads with neutrals of all elements form a circuit, terminals of each branch of all elements form identical circuit, is called balanced (wye) (three-phase) circuit, and the circuit obtain by applying Y-Δ transformation to all wye generator and loads is called balanced (delta) (three-phase) circuit.

XI Transient response

i RC circuit transient response

The response of a capacitor with initial voltage $v(0^-)$ in series connection with a resistance R and step voltage source $v(\infty)u(t)$ is:

$$v(t) = v(\infty) + (v(0^-) - v(\infty))e^{-\frac{t}{\tau}}$$

$$i(t) = \frac{v(\infty) - v(0^-)}{R}e^{-\frac{t}{\tau}}$$

where $\tau = RC$ is the time constant.

ii RL circuit transient response

The response of an inductor with initial current $i(0^-)$ in parallel connection with a resistance R and a step current source $i(\infty)u(t)$ is:

$$i(t) = i(\infty) + (i(0^-) - i(\infty))e^{-\frac{t}{\tau}}$$

$$v(t) = (i(\infty) - i(0^-))Re^{-\frac{t}{\tau}}$$

where $\tau = \dfrac{L}{R}$ is the time constant.

iii Quality factor of second-order circuit

For a second-order circuit with $0 < \zeta < 1$, the energy stored must be in the form

$$E(t) = Ke^{-2\alpha t}\cos^2\left(\omega_d t + \theta\right)$$

$$E_m = Ke^{-2\alpha t_n}$$

$$E_l = Ke^{-2\alpha t_n} - Ke^{-2\alpha(t_n + \frac{2\pi}{\omega_d})} = E_m\left(1 - e^{-\frac{4\pi\alpha}{\omega_d}}\right) = E_m\left(1 - e^{-\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}}\right)$$

$$Q = \frac{2\pi E_m}{E_l} = \frac{2\pi}{1 - e^{-\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}}}$$

For circuit with large quality factor Q, i.e., small ζ, called high-Q or high Q circuit, which we usually take $Q > 5$, i.e., $\zeta < 0.1$, we have $\zeta \approx 0$, and $1 - e^{-x} \approx x$ for $x \approx 0$, thus

$$Q \approx \frac{2\pi}{\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}} = \frac{\sqrt{1-\zeta^2}}{2\zeta} \approx \frac{1}{2\zeta}$$

In most contexts, Q is redefined as $\dfrac{1}{2\zeta}$ for all second order circuits even if they are not high-Q circuit.

XII s-domain analysis

i LTI system

A LTI network is used interchangeably with the causal continuous-time linear time-invariant (LTI) systems with the complementary equation of its governing equation being the governing equation of ZIR and the transfer function $H(s)$ of the system. The transfer function is also called network function. The response is taken $\Re(Y(t))$ to obtain real voltage or current response. The order of the LTI system is the order of the circuit and is the order of the governing ODE.

ii Modes of natural response

Poles determine natural response. Suppose $H(s)$ (with no pole-zero cancellation) has distinct pairs of conjugate poles $\sigma_1 \pm j\omega_1, \dots, \sigma_n \pm j\omega_n$ with multiplicity $m_1, \dots, m_n$. Each of them determines a mode of the natural response

$$y_i(t) = \sum_{j=1}^{m_i} \left|A_{ij}\right| t^{j-1} e^{\sigma_i t} \cos\left(\omega_i t + \operatorname{Arg}\left(A_{ij}\right)\right),$$

where A_{ij} is arbitrary complex constant.

The sum of all modes is the natural response.

$$y_N(t) = \sum_{i=1}^{n} y_i(t).$$

iii Stability

Marginal stable circuit with pure imaginary poles is also called oscillatory or AC circuit, or oscillator.

Marginal stable circuit with no pure imaginary pole is also called DC circuit.

A LTI circuit is Lyapunov stable if but not only if it there is no controlled source in it.

iv Zeros

A zero on the imaginary axis $j\omega_0$ means that sinusoidal input of frequency ω_0 (DC for $\omega_0 = 0$) produces 0 output.

v Pole-zero cancellation

When analyzing forced responses, each multiplicity of a pole and a zero at same place can cancel, while they can't when analyzing natural response since the two are two independent energy-storage elements.

vi DC gain

Low-pass gain is also called DC gain.

vii Resonant rise

In series resonance, the phenomenon that amplitude of voltage of element is larger than that of source is called resonant voltage rise.

In parallel resonance, the phenomenon that amplitude of voltage of element is larger than that of source is called resonant current rise.

viii Cutoff frequency

For a filter with pass gain 1, cutoff frequency is also called half-power frequency since $P_{out} = \frac{1}{2}P_{in}$ at it.

ix Butterworth band-pass and band-stop filter

Transform LP:

$$H(s)\frac{1}{s+1}$$

to BP:

$$H(s) = \frac{1}{\frac{s^2+{\omega_0}^2}{Bs}+1} = \frac{Bs}{s^2+Bs+{\omega_0}^2}$$

to BS:

$$H(s) = \frac{1}{\frac{Bs}{s^2+{\omega_0}^2}+1} = \frac{s^2+{\omega_0}^2}{s^2+Bs+{\omega_0}^2}$$

$$B = 2\alpha = 2\omega_0\zeta = \frac{\omega_0}{Q}$$

$$\omega_H\omega_L = {\omega_0}^2$$

$$\omega_H - \omega_L = B$$

$${\omega_L}^2 + B\omega_L - {\omega_0}^2 = 0$$

$$\omega_L = \frac{-B+\sqrt{B^2+4{\omega_0}^2}}{2}$$

$$\frac{\omega_H+\omega_L}{2} = \omega_L + \frac{B}{2} = \frac{\sqrt{B^2+{\omega_0}^2}}{2} = \omega_0\sqrt{1+\left(\frac{B}{2\omega_0}\right)^2} = \omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2}$$

For high-Q circuit, $\omega_0 \gg B$, called narrow band-pass filter, and

$$\frac{\omega_H+\omega_L}{2} \approx \omega_0$$

Winding resistance of inductors does not significantly affect band-pass filter, but makes the BS filter no ideal but has the form:

$$H(s) = \frac{s^2+2\beta s+{\omega_0}^2}{s^2+Bs+{\omega_0}^2}$$

where $\beta \ll \frac{B}{2}$. If $\beta = 0$, the filter is ideal. Otherwise the filter is called degraded.

$$H(j\omega) = \frac{-\omega^2+2\beta j\omega+{\omega_0}^2}{-\omega^2+Bj\omega+{\omega_0}^2}$$

$$H(j\omega_0) = \frac{2\beta}{B} \ll 1$$

Thus, signal at ω_0 is much reduced but not totally rejected.

x Quadratic factor

$$H_{QD}(s;\omega_0,Q) \coloneqq \frac{{\omega_0}^2}{s^2 + \frac{\omega_0}{Q}s + {\omega_0}^2}$$

- If $Q \geq \frac{1}{2}$, it is a low-pass filter.
- If $Q > \frac{1}{2}$, the maximum of g_{QD} occurs at peak frequency $\omega = \omega_p = \omega_0\sqrt{1 - \frac{1}{2Q^2}}$ and is bigger when Q is bigger, and the natural response is underdamped.
- If $Q = \frac{1}{2}$, at $\omega = \omega_0\sqrt{1 - \frac{1}{2Q^2}} = 0$,

  $$a_{QD}(\omega;\omega_0,Q) = \infty$$

  $$g_{QD}(\omega;\omega,Q) = \infty$$

  and the natural response is critically damped.
- If $Q < \frac{1}{2}$, it is unstable, and the natural response is overdamped.

XIII Transformer

i Ideal transformer

An ideal transformer is a two-port, linear, time-invariant, lossless network consists of two magnetically coupled windings, arbitrarily assigned as primary and secondary coil respectively, where the primary coil has N_1 turns, voltage v_1, and current i_1, and the secondary coil has N_2 turns, voltage v_2, and current i_2. Here, the voltage and current of the two coils are assumed to be measured from opposite directions, that is, dots on the same side. The turn ratio n is defined as

$$n = \frac{N_1}{N_2}.$$

Magnetic coupling of same magnetic flux per turn enforces

$$\frac{v_1}{N_1} = \frac{v_2}{N_2}.$$

Power consistency enforces

$$v_1 i_1 = v_2 i_2.$$

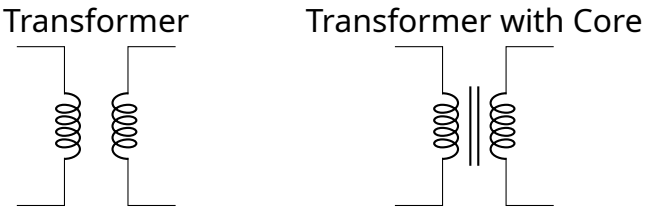
Thus

$$\frac{i_1}{N_2} = \frac{i_2}{N_1}.$$

That is,

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix} \begin{bmatrix} v_2 \\ i_2 \end{bmatrix}.$$

ii Diagram



iii Dot convention

On diagram, a dot is drawn on one terminal of each coil. If dots of the two coils are on the same side, the voltages (and currents) of them as measured in the same direction are in-phase; if dots of the two coils are on the opposite side, the voltages (and currents) of them as measured in the same direction are out-of-phase. Equivalently, if the voltage (and current) of one coil is measured from the dotted terminal to the undotted terminal and the voltage (and current) of the other coil is measured from the undotted terminal to the dotted terminal, the measured voltages (and currents) of them in-phase.

iv Coupled inductors model

Primary and secondary coils are inductors with self inductance L_1 and L_2 and mutual inductance M, that is,

$$v_1 = L_1 \frac{\mathrm{d}i_1}{\mathrm{d}dt} + M \frac{\mathrm{d}i_2}{\mathrm{d}t}$$
$$v_2 = L_2 \frac{\mathrm{d}i_2}{\mathrm{d}dt} + M \frac{\mathrm{d}i_1}{\mathrm{d}t},$$

with constraints that inductive coupling factor is one,

$$\frac{M}{\sqrt{L_1 L_2}} = 1$$

and

$$L_1 = n^2 L_2$$

i.e.,

$$M = nL_2$$
$$L_1 = n^2 L_2$$

i.e.,

$$M = \frac{L_1}{n}$$
$$L_2 = \frac{L_1}{n^2}.$$

This leaves one degree of freedom, thus an arbitrary positive L_1 or L_2 can be assigned.

v VCVS and CCVS model of transformer

Primary coil is replaced with a VCVS

$$v_1 = nv_2.$$

Second coil is replaced with a CCCS

$$i_2 = ni_1.$$

vi Impedance reflection model

Only works if the two coils' circuit are only coupled by the transformer.

If a load impedance Z_s is on secondary coil, seen from primary coil, it's a load impedance

$$Z_p = n^2 Z_s.$$